MIL-T-80131B
10 June 1986
SUPERSEDING
MIL-T-80131A
10 September 1984

MILITARY SPECIFICATION

TURNING ROLLS, WELDING, POWERED TYPE AND IDLER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven turning rolls and mating idler turning rolls for supporting and rotating cylindrical workpieces in welding and related processes.

1.2 Classification. Turning rolls shall be of the following types and sizes. Type I turning rolls shall be powered and have a turning capacity at least three times the support capacity. Type II turning rolls shall be of the idler type. The size of each individual turning roll is designated by the support capacity. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Powered turning rolls.

Size 2 -	6,000 lb. turning	x	2,000 lb.	support capacity.
Size 3 -	10,000 lb. turning	x	3,300 lb.	support capacity.
Size 4 -	15,000 lb. turning	x	5,000 lb.	support capacity.
Size 5 -	20,000 lb. turning	x	7,000 lb.	support capacity.
Size 6 -	30,000 lb. turning	x	10,000 lb.	support capacity.
Size 7 -	40,000 lb. turning	x	13,000 lb.	support capacity.
Size 8 -	50,000 lb. turning	x	17,000 lb.	support capacity.
Size 9 -	60,000 lb. turning	x	20,000 lb.	support capacity.
Size 10 -	75,000 lb. turning	x	25,000 lb.	support capacity.
Size 11 -	100,000 lb. turning	x	34,000 lb.	support capacity.
Size 12 -	120,000 lb. turning	x	40,000 lb.	support capacity.
Size 13 -	150,000 lb. turning	x	50,000 lb.	support capacity.
Size 14 -	180,000 lb. turning	x	60,000 lb.	support capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type II - Idler turning rolls.

Size 2	-	2,000 lb. support capacity.
Size 3	-	3,300 lb. support capacity.
Size 4	-	5,000 lb. support capacity.
Size 5	-	7,000 lb. support capacity.
Size 6	-	10,000 lb. support capacity.
Size 7	-	13,000 lb. support capacity.
Size 8	-	17,000 lb. support capacity.
Size 9	-	20,000 lb. support capacity.
Size 10	-	25,000 lb. support capacity.
Size 11	-	34,000 lb. support capacity.
Size 12	-	40,000 lb. support capacity.
Size 13	-	50,000 lb. support capacity.
Size 14	-	60,000 lb. support capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NEMA ICS 1 - Industrial Control and Systems.

(Application for copies should be addressed the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to The National Fire Protection Association, Batterymarch Park, Quincy, MA 02269).

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The turning rolls shall be new and one of the manufacturer's current models designed so that one power driven turning roll and one or more mating idler turning rolls shall support and rotate a cylindrical workpiece of the size and capacity specified herein. The turning rolls shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified but are a part of the manufacturer's current design shall be included in the turning rolls furnished. When specified (see 6.2.1), the turning rolls shall be designed with the capability of rotating workpieces with unusual configurations under unbalanced load conditions. Turning roll parts and components subject to contamination from flux and other foreign matter shall be enclosed. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Type I turning rolls shall be power driven. Type II turning rolls shall be of the same design as type I, except that they shall not be of the power driven type.

3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the turning rolls. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The turning rolls may contain reclaimed materials provided such materials will not jeopardize their intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the turning rolls.

3.2.3 Energy efficiency. The turning rolls and any components which directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

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3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the turning rolls that present safety hazards. The safety devices shall not interfere with the operation of the turning rolls. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the turning rolls, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the turning roll itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The turning rolls shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the turning rolls.

3.2.8 Environmental protection. The turning rolls shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the turning rolls shall not emit materials hazardous to the ecological system as prohibited by federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or turning rolls.

3.3 Construction. The turning rolls shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the turning roll at its maximum rating and capacity without permanent distortion.

3.3.1 Casting and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the turning roll. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the turning rolls shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the turning roll and its attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The major component of all turning rolls shall be a chassis assembly, which will support two or more turning roll wheel assemblies. The powered turning roll chassis assembly shall also support a wheel drive assembly and controls.

3.4.1 Chassis assembly. The chassis assembly shall be a steel weldment of rigid construction that will maintain original alignment and minimize distortion. The base of the chassis shall provide for bolting the chasis to the floor. Size 3 and larger turning rolls shall be provided with necessary wheel adjustment to accomodate workpieces of different diameters within the range specified in table I. All shafts and guards required for operation of the turning rolls in all adjustment positions shall be provided. All wheels on a common chassis shall be aligned to such a degree as to assure maximum contact between the wheel surfaces and the workpiece.

3.4.2 Wheel assemblies. Wheel assemblies shall consist of one or more pedestals, axles, bearings, and wheels. These items shall be of such material, size, quantity, and load carrying ability as required to meet all performance requirements of the turning rolls, including the loading, turning, and unloading of the workpiece. Unless otherwise specified, wheels shall have rubber tired surfaces. When specified (see 6.2.1), wheels shall have non-ferrous or steel tired surfaces. Unless otherwise specified, (see 6.2.1), the wheel tire diameter and width shall be the manufacturer's

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standard. Rubber tired wheels shall be provided with a means for tire replacement. When specified (see 6.2.1), rubber tired wheels shall be provided with steel overload protection rings.

3.4.3 Wheel drive system. Type I turning rolls shall be provided with an electric motor drive system. The system shall provide not less than the turning capacity specified in table I. Drive mechanisms shall be enclosed, where feasible, and shall be properly lubricated for continuous performance at rated loads. The drive system shall provide a variable surface speed to the driven wheels, with a range as specified in table I. The drive system shall hold the workpiece in a stationary position in the event of power loss.

3.4.4 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Size 3 and larger powered turning rolls shall operate from a single 230/460 volt, 3 phase, 60 Hz circuit. Each power operated turning roll shall have a fused safety disconnect switch or circuit breaker. Unless otherwise specified (see 6.2.1), each powered turning roll size 3 and larger shall be initially wired for operation on 460 volts. Size 2 power operated turning rolls shall operate on a 120 volt, 60 Hz, single phase power supply. An identified terminal for grounding the turning roll shall be mounted on or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. Electronic type drive controls shall be mounted in an enclosure conforming to ANSI/NEMA ICS 1.

3.4.4.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. The turning rolls shall be equipped with a reversing starter for operation in either direction. The reversing starter shall be provided with thermal overload protection. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure conforming to ANSI/NEMA MG 1.

3.4.4.2 Controls. Controls for operating the powered turning rolls shall be assembled in a pendant type enclosure mounted at the end of a multi-conductor control cable not less than 20 feet long. When specified (see 6.2.1), the cable attachment shall be a reel type. The control shall include forward, reverse, and stop pushbuttons with a "power on" pilot light. On mechanical drive units, the variable speed control shall be mounted on the drive motor unit. When required (see 6.2.1), the control system shall be furnished with a foot switch, or other remote control device, as specified. The control circuits shall operate on 120 volts.

3.4.4.3 Solid state components. Solid state design shall be used throughout for electronic components except on mechanical drive systems. The electrical system shall have a standard mechanical starter, fused disconnect, and transformer.

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TABLE I. Turning roll capacities.

Turning roll size	Turning capacity (pounds) type I only	Support capacity (pounds) type I and II	Motor horse-power type I only minimum	Roll wheel surface speed (inches per minute)		Workpiece diameter	
				minimum	maximum	minimum (inches)	maximum (feet)
2	6,000	2,000	1/3	2	60	3	3
3	10,000	3,300	1/2	2	60	4	12
4	15,000	5,000	1/2	2	80	4	13
5	20,000	7,000	1/2	2	80	4	13
6	30,000	10,000	3/4	2	80	4	13
7	40,000	13,000	3/4	2	80	4	13
8	50,000	17,000	1	2	80	5	20
9	60,000	20,000	1-1/2	2	64	5	20
10	75,000	25,000	2	2	64	8	20
11	100,000	34,000	2	2	64	8	20
12	120,000	40,000	3	2	64	8	20
13	150,000	50,000	5	2	64	14	21
14	180,000	60,000	5	2	64	8	18

NOTE: All requirements shall be not less than the figure shown except that where a "minimum" is shown the requirement shall be not more than the figure shown, and where a "maximum" is shown the requirement shall be not less than the figure shown.

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3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the capacities of the turning rolls shall be not less than those specified in table I for the turning roll size specified.

3.6 Performance. Type I turning rolls shall be capable of rotating a cylindrical workpiece having the total weight specified in table I. The speed shall be variable, stepless, and reversible, and the wheel surface speed shall be accurate to within ±3 percent of selected speed. Type II turning rolls shall be capable of supporting not less than the weight specified in table I. Each turning roll shall be capable of withstanding the normal rough usage encountered by this type equipment during loading, turning, and unloading workpieces. Each turning roll shall accept workpieces having diameters within the range specified in table I.

3.7 Standard equipment. Unless otherwise specified, all standard equipment normally furnished by the manufacturer shall be furnished. Additional equipment required shall be furnished as required and fully described (see 6.2.1).

3.8 Optional equipment. Optional equipment specified in the following sub-paragraphs shall be furnished with each turning roll when specified (see 6.2.1).

a. Magnetic brakes or electric motor brakes.

b. Tachometer (accurate to within ±2 percent at maximum indication).

c. Rotary type grounding device.

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each turning roll. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each turning roll. The nameplate shall contain the information listed below. If the turning roll is a special model,

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the model designation shall include the model of the basic standard turning roll and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Turning capacity, pounds
Support capacity, pounds
Workpiece diameter, minimum and maximum
Manufacturer's name
Manufacturer's model designation
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the turning rolls and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to tne first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first acticle inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The turning rolls shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The type I turning rolls shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period. The type II turning rolls shall be manually operated for a period of not less than 5 minutes to determine proper operation of all components.

4.7.2 Performance test. Each turning roll shall be tested for conformance to the performance requirements of 3.6. An actual or simulated supported load shall be applied to each turning roll. A turning load shall be applied to each powered roll. The load shall be equal to the maximum load specified in table I for the size unit being tested.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

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5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. Turning rolls described by this specification are intended for use in welding shops to rotate cylindrical workpieces while welding in a general downward style. The rolls may be used in automatic and precision welding applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. Capability of handling workpieces with unusual configurations under unbalanced load conditions, if required (see 3.2).

e. If turning rolls are required to be configured in a specific dimensional system (US or SI), state required system (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

i. Wheel tire surfaces, if different (see 3.4.2).

j. Wheel tire diameter and width, if different (see 3.4.2).

k. Wheel protection rings, if required (see 3.4.2).

l. Electrical system and input power, if different (see 3.4.4).

m. Motor enclosure, if different (see 3.4.4.1).

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n. Reel type cable and foot controls or other remote controls, if required (see 3.4.4.2).

o. Size and capacity, if different (see 3.5).

p. Specify and describe standard equipment, if different (see 3.7).

q. Specify and fully describe optional equipment, if required (see 3.8).

r. Specify applicable repair parts, if required (see 3.9).

s. Lubrication chart or plate, if different (see 3.10.1).

t. Nameplate, if different (see 3.10.2).

u. First article inspection, if different (see 4.4).

v. Quality conformance inspection, if different (see 4.5).

w. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports, should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the turning roll itself. It does not require compliance with those OSHA requirements that concern the turning roll in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OHSA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual turning rolls in the environment, the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirements that will integrate the new turning roll into its future operating environment. The turning roll shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial turning roll supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity

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should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the work-pieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(f).

6.5 Subject term (key word) listing.

Cylindrical workpiece welding
Idler rolls, welding
Rotational workpiece welding
Support rolls, welding
Turning rolls, welding
Welding, cylindrical workpieces

6.6 Cross reference of classification changes. Turning roll classification changes from the previous issues of this specification are as follows:

MIL-T-80131	MIL-T-80131A	MIL-T-80131B
Type I	(No change)	(No change)
Type II	(No change)	(No change)
Size 1	Deleted	
Size 2	(No change)	(No change)
Size 3	(No change)	(No change)
Size 4	(No change)	(No change)
Size 5	(No change)	(No change)
Size 6	(No change)	(No change)
Size 7	(No change)	(No change)
Size 8	(No change)	(No change)
Size 9	(No change)	(No change)
Size 10	(No change)	(No change)
Size 11	(No change)	(No change)
		Size 12-(New size)
		Size 13-(New size)
		Size 14-(New size)

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activity:
Navy - MC

Preparing activity:
DLA - IP

(Project 3436-0035)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-80131B	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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